MMAE 414 Final Report

Sky Combine - Group 7

December 1st, 2025

Kaleb Martiny – Group Lead, 2nd Aerodynamics

Gavin Coffer – Cost Analysis Lead, 2nd Structures

Alec Weedman – Propulsion Lead, 2nd Group Lead

Logan Garcia – Aerodynamics Lead, 2nd Cost Analysis

Leah Popoolapade – Structures Lead, 2nd Propulsion

Contents

1. Introduction

The **Sky Combine** is an agricultural aircraft designed to compete with current aircraft in the market while reducing the threat to human life and property while optimizing operational processes of crop-dusting. Through conceptual design reviews, trade studies, and flight simulations, the Sky Combine was optimized to be able to complete its dusting missions. A model of the aircraft was developed in XFLR5 where aerodynamic forces were calculated as the model underwent analysis' through different flight conditions and control surface deflections. These aerodynamic values were crucial in being able to run flight simulations in MATLAB and Simulink.

The aircraft must complete different maneuvers to be able to effectively operate in its crop-dusting mission. The four maneuvers that the Sky Combine is shown to complete in this report are: Glide Performance, Takeoff/Climb Performance, Holding Performance, and Landing Performance. First, the Glide Performance must demonstrate the aircraft in un-powered gliding flight to determine its glide trajectory at trim conditions. Second, Takeoff and Climb Performance involve modeling the aircraft's climb rate as it takes off and prepares its crop-dusting mission. Next, the Holding Performance involves the aircraft holding a fix about a waypoint and then completing a turn as it sprays crops until it returns to its initial position. Finally, the Landing Performance involves simulating the aircraft preparing for landing.

2. XFLR5 Data and Coefficient Graphs

Before being able to use Simulink for flight simulation, parameters from the Sky Combine were calculated and gathered from XFLR5 shown in the table below. These parameters are precise values from the aircraft that were used to generate aerodynamic coefficients when the aircraft's control surfaces were changed or when the aircraft experienced a disturbance such as sideslip.

Variable	Value	Unit
Mass	7903.82	kg
X_{cg}	-1.065	m
X_{np}	-2.022	m
Span	23.5	m
Reference Area	70.50	m^2
Cruise Speed	92	m/s

Table 1: XFLR5 Parameters of the Sky Combine

Additionally, XFLR5 gave calculated values of the inertia matrix of the Sky Combine model:

$$\mathbf{I} = \begin{bmatrix} 57312.99 & 0 & 690.73 \\ 0 & 12860.02 & 0 \\ 690.73 & 0 & 70001.52 \end{bmatrix}$$ (kg*m^2)

To be able to compute flight simulations, using MATLAB and Simulink, matrices for the datum and increments of the surface deflections (rudder, aileron, and elevator) and the side force effect (side slip angle) are needed to determine the aerodynamic coefficients of the aircraft.

The datum can be represented when the aircraft is on level flight at cruise conditions with no surface deflections. This was used to gather a general representation of the aerodynamic forces at trim conditions and can be observed in **Figures 1.1-1.3: CL, CD, Cm vs. Angle of Attack**. A trim angle of 1.91 degrees was calculated where Cm=0 at a positive angle of attack and dCm/dalpha < 0, which allows a conclusion that the aircraft has longitudinal static stability. The lift coefficient of the aircraft is observed to increase as angle of attack increases due to the aircraft's ability to generate higher lift as the nose is angled higher. The drag coefficient also increases as the angle of attack increases. It approaches zero from lower angles of attack and then begins to increase at higher angles.

Additionally, using the datum configuration of the aircraft, elevator deflection was used to determine aerodynamic coefficients of lift, drag, and pitching moment which was plotted against angle of attack. These are represented in **Figures 2.1-2.3: dCL, dCD, dCM for Elevator Deflection**. Longitudinal static stability is also confirmed when there is positive elevator deflection (trailing edge down) and there is a positive moment coefficient, and vice versa for a negative deflection (trailing edge up). This can be observed below in **Figure 2.3** for the pitching moment coefficient with elevator deflection.

For the lateral-directional stability of the aircraft to be computed, the surface deflections of the rudder, aileron, and the side slip angle are used. Rudders primarily control the yaw moment; ailerons primarily control the rolling moment; and side slip angle primarily controls the side force. These deflections influence the coefficients of yaw, roll, and side force which are Cn, Cl, and Cy, respectively. These are represented in: **Figures 5.1-5.3: dCY, dCl, dCn for Rudder Deflection, Figures 4.1-4.3: dCY, dCl, dCn for Aileron Deflection, and Figures 3.1-3.3: dCY, dCl, dCn for Beta Side Slip.** It can be observed in the graphs that the aircraft has lateral-directional static stability such that: when there is positive aileron deflection it produces a positive rolling moment Cl, and vice versa for negative; when there is a positive rudder deflection it produces a negative yawing moment Cn, and vice versa; and a positive side slip produces a negative Cl that rolls away from the side slip angle.

Figures 1.1-1.3 CL, CD, Cm vs. Angle of Attack

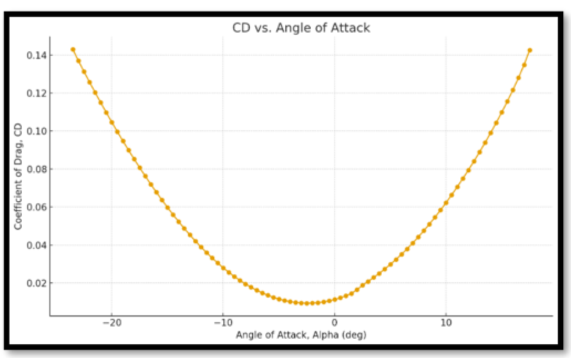


Figure 1.1 CL vs. AOA

Figures 2.1-2.3 dCL, dCD, dCM for Elevator Deflection

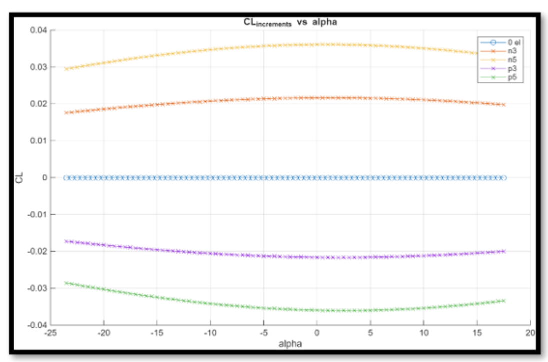


Figure 2.1 dCL Elevator

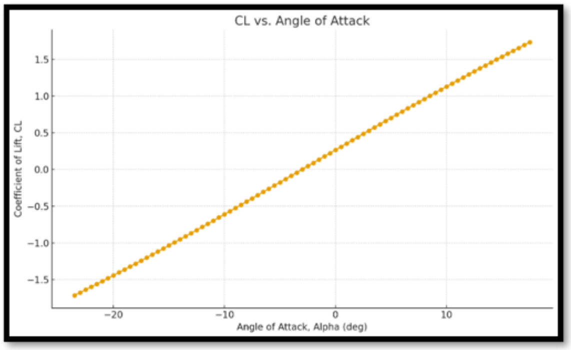


Figure 1.2 CD vs. AOA

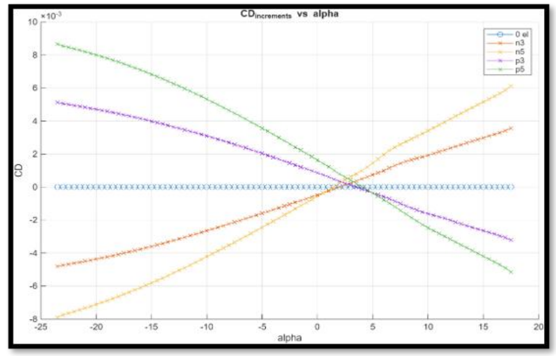


Figure 2.2 dCD Elevator

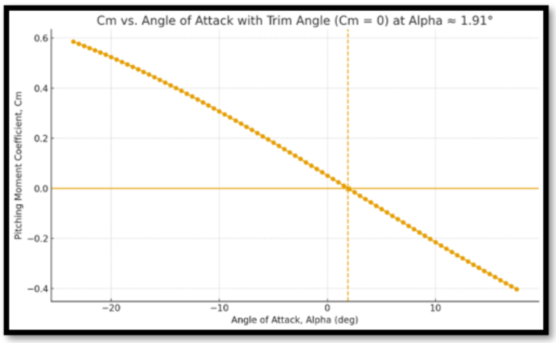


Figure 1.3 Cm vs. AOA

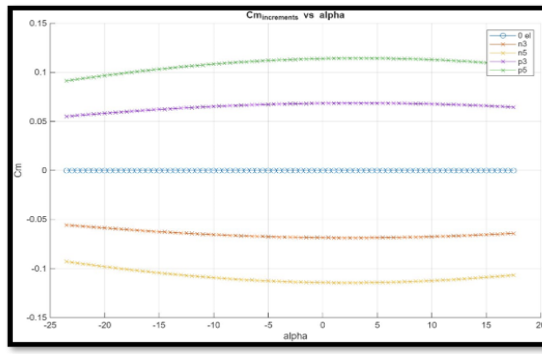


Figure 2.3 dCm Elevator

Figures 3.1-3.3: dCY, dCl, dCn for Beta Side Slip

Figures 4.1-4.3: dCY, dCl, dCn for Aileron Deflection

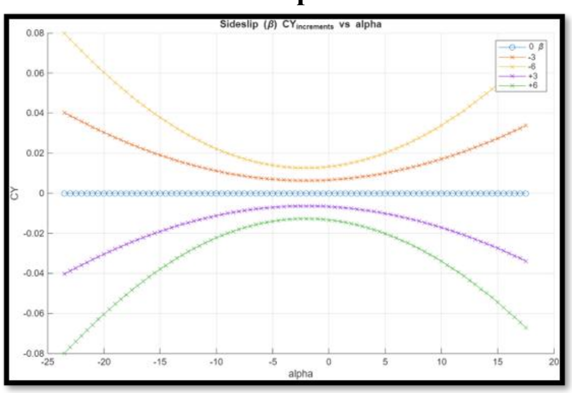


Figure 3.1 dCY Beta

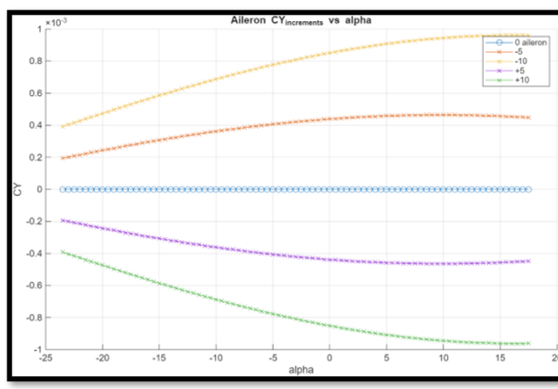


Figure 4.1 dCY Aileron

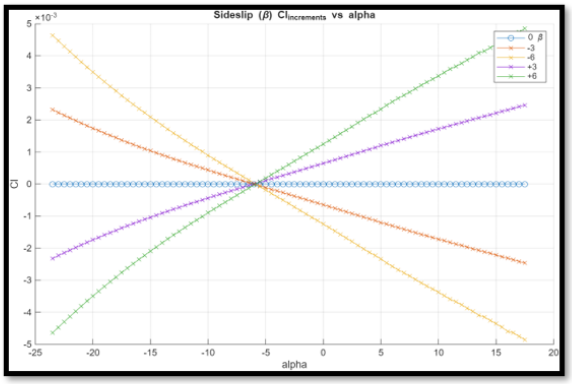


Figure 3.2 dCl Beta

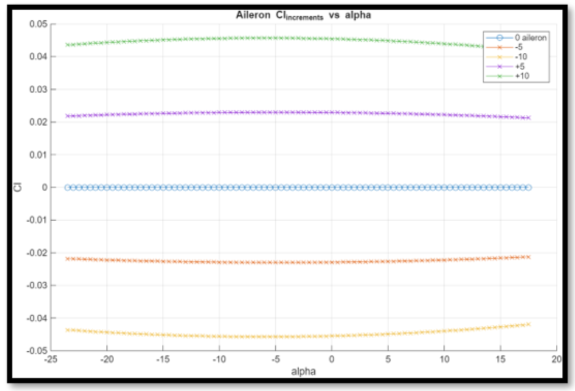


Figure 4.2 dCl Aileron

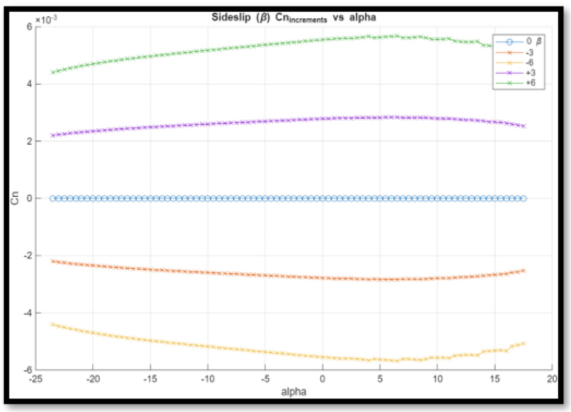


Figure 3.3 dCn Beta

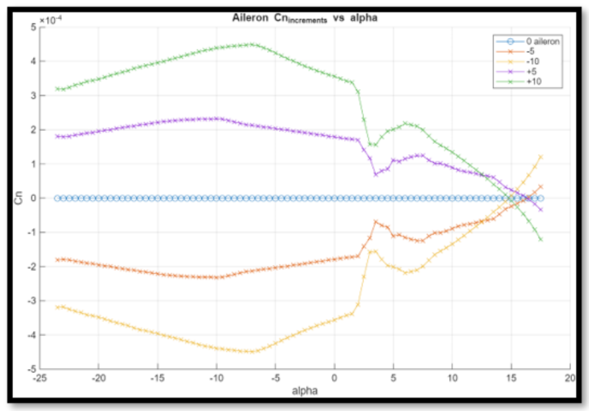


Figure 4.3 dCn Aileron

Figures 5.1-5.3: dCY, dCl, dCn for Rudder Deflection

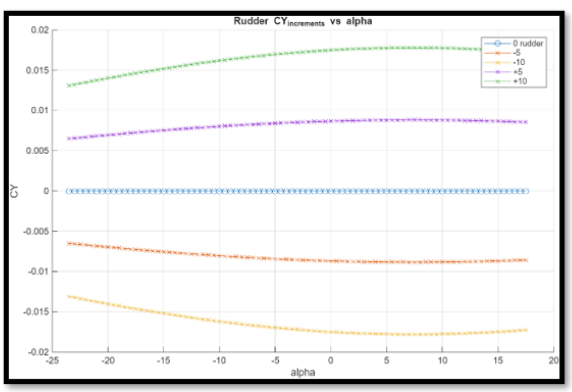


Figure 5.1 dCY Rudder

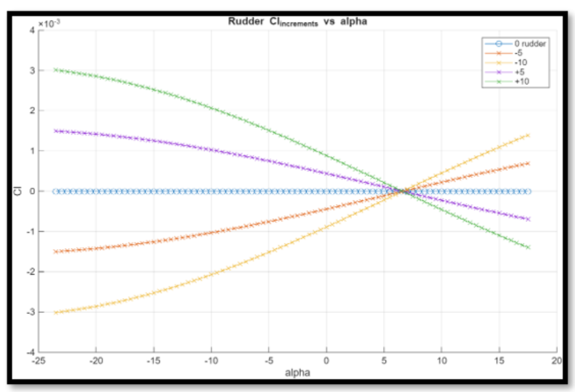


Figure 5.2 dCl Rudder

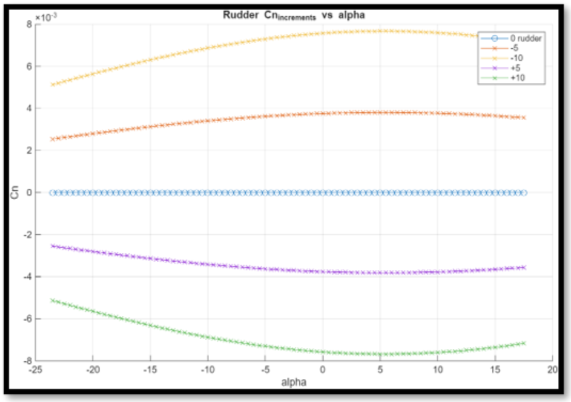


Figure 5.3 dCn Rudder

3. Simulink Control Blocks

Simulink was used to create a 6 degree of freedom model to simulate flight maneuvers of the aircraft. PID controllers were used to try and reduce the settling time and percentage overshoot of the aircraft system. The Simulink Model found in Appendix 1 shows the overall model used for the Sky Combine aircraft.

4. Glide Performance

This simulation requires the aircraft to achieve unpowered gliding flight from a starting elevation of 5,000 feet AGL under trim conditions. The result of this simulation is to determine the glide trajectory of the aircraft, quantify its glide ratio, and compare this to the data calculated by xflr5 at the trim condition.

Control Surface	**Ailerons**	**Elevator**	**Rudder**
Deflection Angle (Degrees)	**0**	**1.913**	**0**
Gliding Flight Airspeed	**80.75 m/s**	**264.9 ft/s**	**156.9 knots**
CL, CD, L/D at Trim	**CL = 0.4336**	**CD = 0.0161**	**L/D = 26.9**

Table 2: Tabulation of the aircrafts control surface trim conditions, trim air speed, and lift and drag coefficients at the trim condition

The ailerons and rudder experience no deflection at trim, as the aircraft is symmetrical across the x-z plane. The elevator requires a deflection of 1.913 degrees to achieve static stability as seen in Fig. 1.3. These values are important for all future simulations because they define the equilibrium point of the aircraft. When paired with thrust in later simulations, this will result in steady, level, unaccelerated flight.

The initial conditions for this simulation are an initial altitude of 5,000 feet AGL, no thrust, and the controls fixed at their trim conditions defined in Table 2. The Simulink model used can be found in Section 3. With the elevator deflection set to a constant 1.913 degrees, a phugoid damping feedback loop was used to control the overall velocity of the descent. The velocity was adjusted until the output elevator deflection was a near constant 1.913 degrees across the entire simulation. The resulting velocity that achieved this was 80.75 m/s, or 156.9 knots.

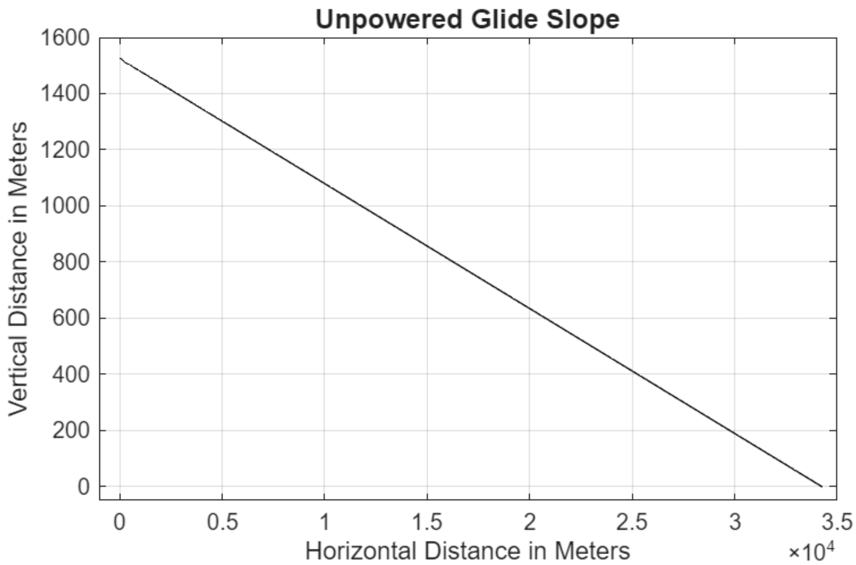


Figure 6: Powered off glide distance

Figure 6 is the resulting simulated glide trajectory of the aircraft at trim conditions. The downward slope of the graph is expected and is consistent with that of gliding flight. There is a small amount of phugoid oscillation at the beginning of the glide slope, however this is quickly damped out at the aircraft reaches equilibrium. To reduce phugoid oscillations as much as possible beyond the addition of a feedback controller, the initial velocity for the simulation matches the final gliding velocity of 80.75 m/s (156.9 knots). The total distance covered during this simulation is $34{,}235\ m = 112319\ ft = 18.50\ nmi$. This was achieved over a simulation time of 425 seconds. Taking the change in horizontal distance divided by the change in vertical distance yields the simulated lift over drag ratio. Using the total horizontal distance covered and the starting height, a lift over drag ratio of 22.46 is achieved. Comparing this to the data calculated in xflr5 found in Table 2, there is some difference between the simulation and what xflr5 calculated. There is one main reason for this difference, which lies in the way xflr5 computes its numbers versus the equations of motion found in the Simulink model. Looking at the way xflr5 computes the coefficients across a 3D wing, it assumes ideal flow under constant, steady conditions. There is no acceleration or phugoid motion present to alter the lift or drag. This contrasts with Simulink, where real flight dynamics come into play. The lift over drag ratio taken from the simulation is not an instantaneous measurement, but instead an average. It is possible that for some moments the ratios matched, but over the course of the glide this was damped out. The lift over drag ratio can also be compared to the following equation, which uses the wing aspect ratio, Oswald efficiency, and zero lift drag coefficient.

$$\left(\frac{L}{D}\right)_{Max} = \frac{1}{2}\sqrt{\frac{\pi A e}{C_{D_o}}} = \frac{1}{2}\sqrt{\frac{\pi * 7.83 * 0.8(or\ 0.96)}{0.0095}} = 22.78(or\ 24.96)$$

The values found in this equation were pulled from xflr5 and [4]. An Oswald efficiency was also calculated from the xflr5 data and can be seen in parentheses along with the resulting ratio. Taking an Oswald efficiency of 0.8, which it was found to be reasonable for this class of aircraft, results in a lift over drag ratio that is only ~1% larger than the value simulated.

5. Level Flight and Climb Performance

This simulation requires the aircraft to fly at 1,500 feet AGL for 10 seconds, after which it will begin to climb to an altitude of 3,500 feet AGL. Once that height is reached, the aircraft will level out and fly for 20 seconds at 3,500 feet AGL. If variable thrust is used, it cannot exceed the maximum output of the engine. The attack angle cannot exceed 15 degrees at any point, and the elevator deflection angle can never exceed 25 degrees in either direction. The simulation shows that the aircraft can complete both level flight, as well as steady climbing flight to a desired altitude, after which level flight can be achieved again.

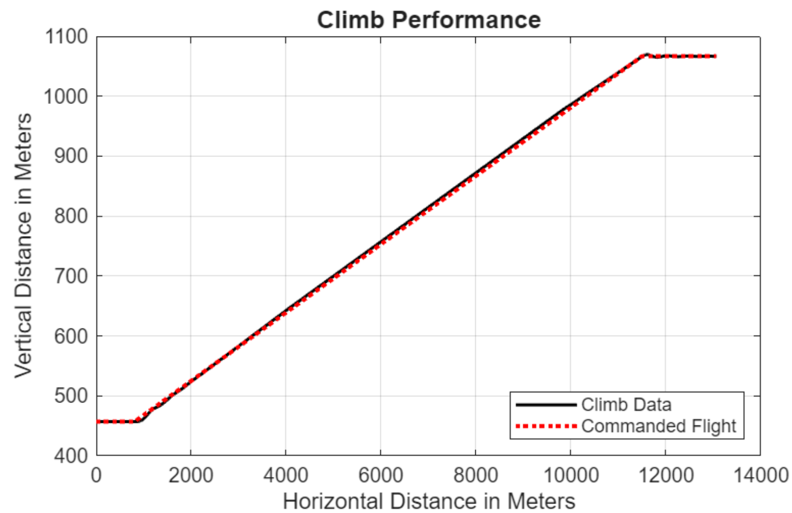


Figure 7: Climb performance

Figure 7 was the result of this simulation, including both the commanded flight and simulated flight. The aircraft was able to very closely follow the commanded flight path, with small deviations during the start and end of the climb. At the start, there was a little bit of lag as the aircraft rotates into climbing flight and settles into stable climb. As seen in Figure 10, this was experienced as phugoid oscillations which are mostly damped out during stable climb, only to return during the level out at the top of the climb. It was at this point where the most deviation from the commanded flight path was seen. At its peak, there was an overshoot of about ten feet, which quickly damped as the aircraft returned to level flight at 3,500 feet. Beyond this point, the aircraft maintained level flight at 3,499 feet, well within margin of error.

The climb rate the aircraft experienced was 905.5 feet per minute. This was calculated by taking two points along the climb trajectory and using the difference in height and time to get feet per minute. The points chosen were (12, 460.1) and (143, 1063.4), with the first number being time and the second being meters. For comparison, the Air Tractor 802A, which the initial design of the aircraft was based on, has a climb rate of 780 feet per minute at 16,000 pounds, giving out aircraft a ~120 feet per minute advantage over the 802A while also being heavier at 17,424 pounds. The time to climb from 1,500 to 3,500 feet was 133 seconds, with the total flight time to complete the simulation being 163 seconds. Looking at Figure 8, the limiting factor for the climb was the available thrust, at least for the given climb velocity of 80.75 meters per second. If the velocity were lowered, the aircraft would be able to climb at a faster rate. Another limiting factor could be the weight of the aircraft. Looking at Figures 9 and 10, the commanded elevator deflection was never reached, indicating that either thrust was lacking, or velocity was too high. In future iterations, this balance could be further optimized.

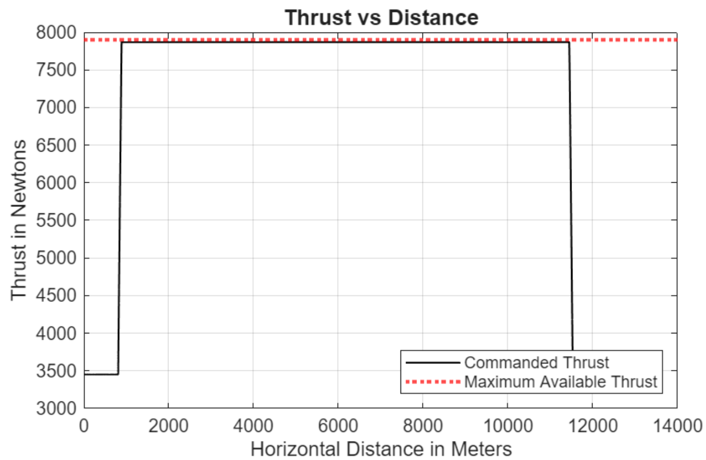


Figure 8: Commanded thrust over horizontal distance

As seen in Figure 8, once the aircraft enters the climb, there is a huge increase in power, right up to the available limit of the engine. This increase was necessary to maintain a velocity of 80.75 meters per second. If a lower velocity were used, less power would be needed. One thing that was not accounted for in this control was the potential ramping up of the engine. The change from thrust at level flight to climbing flight was assumed to happen over the course of one second, however the actual response of the engine may differ from this. This would alter the entrance and exit of the climb.

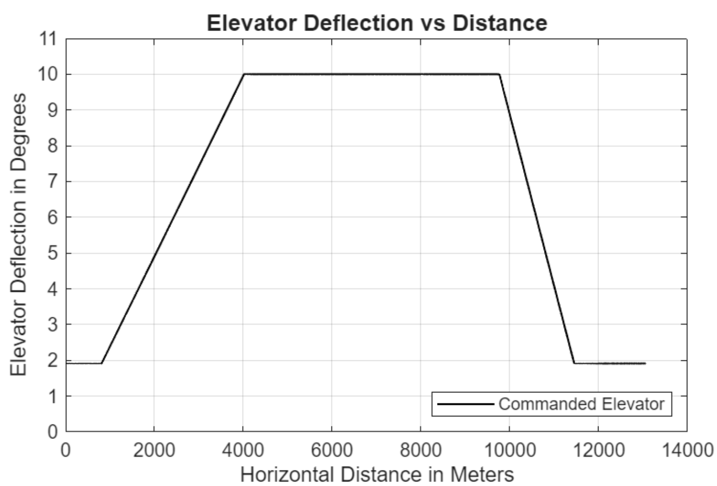


Figure 9: Commanded elevator deflection over horizontal distance

As seen in Figure 9, the commanded elevator deflection was not constant during the initial and final segments of the climb. This was to help reduce sudden changes in angle of attack. When looking at an aircraft entering a climb, unless it is a military jet with very high power to weight ratios and oversized control surfaces, there is almost always a somewhat parabolic shape, rather than a sharp turn, as the aircraft needs to shift its velocity vector to include a new upward component. If this were to happen instantly, not only could it be destabilizing, but it would be uncomfortable for both the pilot and any passengers. Because of this, a sloped approach to the elevator command was taken.

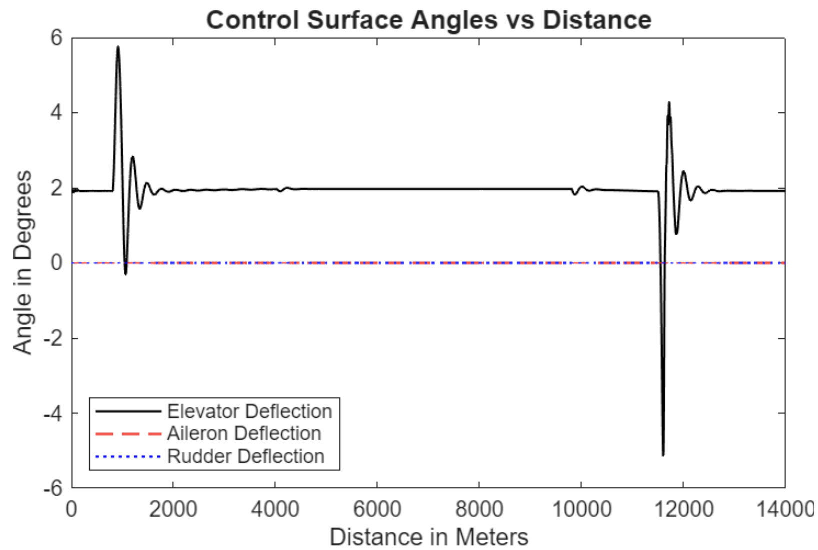
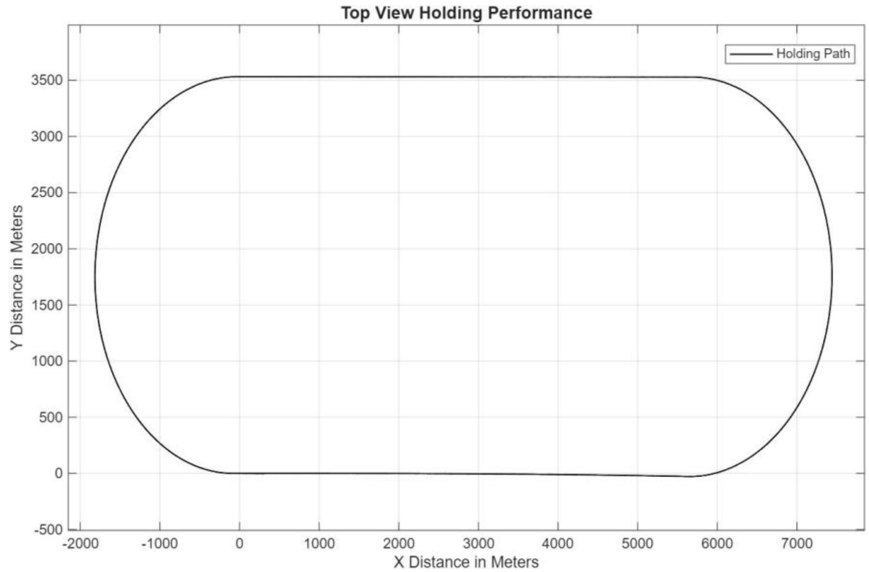


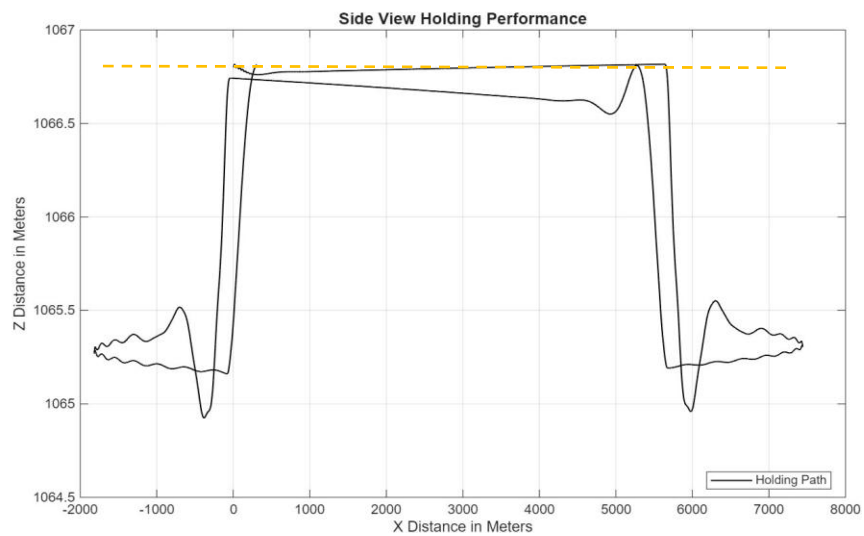
Figure 10: All control surface deflections over horizontal distance

Figure 10 includes the deflection angles of all three control surfaces. The aileron and rudder experience no deflection as expected, and the elevator only experiences deflection during the transition period at the beginning and end of the flight before returning to the trimmed flight deflection of 1.913 degrees.

6. Holding Performance

This simulation requires the aircraft to complete a holding fix about a waypoint for one minute, complete a full 180-degree left turn, hold fix for another minute, and finally finish by completing another 180-degree turn to return at the starting point. Starting at a height of 1066.8 meters (3500ft) above ground level (AGL) at an initial cruise velocity of 92 m/s, Sky Combine heads straight for 5850 meters and begins a ~2.94 deg/s turn for around 62 seconds at a bank angle of 25.65 degrees. The aircraft holds a fixed heading 180 degrees opposite of the initial heading for the same distance and begins the second left turn to make its way back to the initial position. The second turn began with a bank angle of 98% of the first turn and gradually increases till a 25.65-degree bank angle is met. This is due to a change in the altitude of the first turn compared to the second turn as seen in the side view (X, Z) plot. Utilizing only altitude and roll control along with thrust signal inputs, Sky Combine was able to return to the starting point with less than 0.5 m (1.64 ft) deviation in the Y-plane (Top View) and returned to the starting altitude of 1066.8 meters AGL. For better altitude hold performance during the turns, a thrust controller would help provide increased power to maintain the height and velocity of the plane coming out of the straight flight paths. A phugoid damping system would also decrease oscillation during the turn, providing a smoother overall performance for the aircraft. Since the aircraft was able to perform within the required bounds without these controllers and damping systems, they were not needed in these parts.

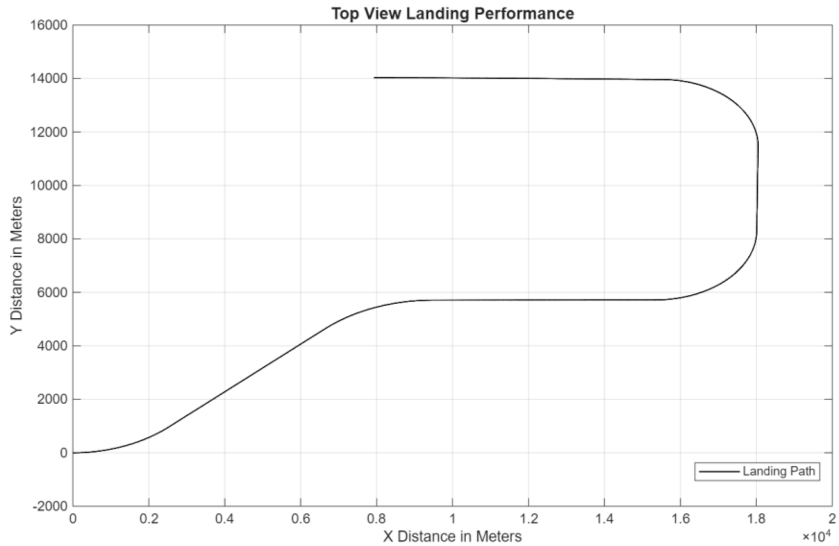


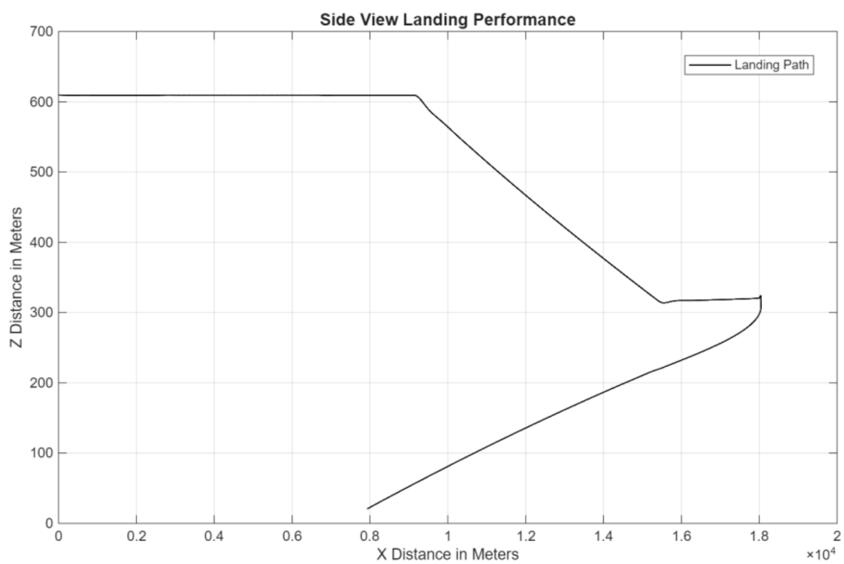


Figures 11.1 & 11.2: Top and Side View of Holding Performance

7. Landing Performance

After initiating a 42-degree approach-angle to a downwind leg, Sky Combine flies along this path for 60 seconds. With an altitude of 606.9 meters (2000 ft) and a cruise speed of 92 m/s, the aircraft enters the downwind leg, completing a right turn to straighten the flight path back to traveling along the x-axis. During the downwind leg, the aircraft is signaled to decrease altitude to 304.8 meters (1000 ft) AGL but increases air speed as it travels downward. A thrust controller incorporated in the simulation model was attempted but unsuccessful in decreasing the velocity properly while also decreasing the altitude during this downwind leg. At a decent angle of 2.7 degrees, traveling vertically downward at 5 m/s. Entering the first 90-degree left turn, like the holding performance, executes the turn with a bank angle of 25.65-degrees for 36.28 seconds. During the base leg, the aircraft altitude drops twenty meters and executes the second 90-degree left hand turn. Exiting this maneuver, the aircraft enters a zero-thrust glide with a decent angle of 1.6 degrees and reaches the ground after a little over 10000-meter horizontal travel. The model utilized the same altitude and roll controllers from the holding performance along with a phugoid damping system. All signal inputs are different and specific to this landing pattern. With this current design, the plane approaches the ground at a velocity of 89 m/s, which would not be considered safe but with the low approach angle it can be done with a large amount of skill. A more ideal approach velocity would be around 60 m/s going into the first 90-degree turn and holding till decent to ground level where the aircraft should reach stall speed to land safely.





Figures 12.1 & 12.2: Top and Side View of Landing Performance

8. Risk Analysis

Risk analysis serves as a fundamental part in the development of any aircraft design, providing a systematic approach to identify, assess, and mitigate potential hazards that could compromise mission success, safety, or compliance. For SHOTS Agricultural Aircraft, this analysis is particularly critical given the inherently high-risk operational environment of agricultural aviation. Furthermore, this analysis provides stakeholders, including potential operators, insurers, and regulatory authorities, with confidence that safety considerations have been addressed throughout the design process.

Risk analysis for this aircraft established industry standards, primarily the Department of Defense MIL-STD-882E System Safety Standard, which provides a structured eight step approach to identifying and managing risks throughout the entire system lifecycle. The analysis framework employs a systematic approach that categorizes risks based on their likelihood of occurrence (ranging from “Frequent” to “Improbable”) and their potential consequences (from “Catastrophic” to “Negligible”). This assessment enables the assignment of Risk Assessment Codes (RAC) that prioritize mitigation efforts and establish appropriate levels of management approval for risk acceptance. For agricultural aircraft operations, this methodology must account for unique operational characteristics including low-altitude maneuvering, remote area operations, and the high operational tempo demanded by seasonal agricultural cycles. The resulting risk matrix provides a clear framework for design decisions, operational procedures, and safety system requirements that will enable the SHOTS aircraft to achieve superior safety performance while maintaining the operational effectiveness essential for commercial agricultural aviation success.

Risk Definitions (MIL-STD-882E)

Risk Probability	Description
Frequent (A)	Often occurs, and is consistent threat to mission
Probable (B)	Will occur several times in the life of an item
Occasional (C)	Likely to occur in the life span
Remote (D)	Unlikely but possible
Improbable (E)	Highly unlikely to occur
Eliminated (F)	Incapable of occurrence

Risk Severity	Description
Catastrophic (1)	Could result in complete mission failure, death, irreversible impact
Critical (2)	Results in partial disability, injuries, environmental impact, or monetary loss.
Marginal (3)	Injury or occupational illness for one or more lost work, moderate environmental impact
Negligible (4)	Injury or occupational illness not resulting in a lost work, minimal environmental impact

SHOTS Risk Assessment Matrix

		Risk Severity			
		Catastrophic (1)	Critical (2)	Marginal (3)	Negligible (4)
Risk Probability	Frequent (A)	1 (High Risk)	2 (High Risk)	7 (Serious Risk)	13 (Medium Risk)
	Probable (B)	3 (High Risk)	5 (High Risk)	9 (Serious Risk)	16 (Medium Risk)
	Occasional(C)	4 (High Risk)	6 (Serious Risk)	11 (Medium Risk)	18 (Low Risk)
	Remote (D)	8 (Serious Risk)	10 (Medium Risk)	14 (Medium Risk)	19 (Low Risk)
	Improbable (E)	12 (Medium Risk)	15 (Medium Risk)	17 (Low Risk)	20 (Low Risk)
	Eliminated (F)	Eliminated			

Detailed Risk Analysis

Hazard	Assessment	Risk	Mitigation Strategy
Pre-Flight			
Manufacturing Quality Control	C2	Structural defects, engine installation errors, and avionics integration failures can stop the aircraft from flight	Enhance quality control and inspection procedures
Ground Support Equipment Failure	C3	Improper fuel delivery can cause delays and engine failures, ground power equipment can damage avionics	Standardized ground support equipment design, redundant loading systems, operator training programs
Maintenance Training Deficiency	B2	Improper maintenance can cause component failures and missing logs can hide existing problems	Comprehensive maintenance manuals, certified technician training, routine inspection protocols
In-Flight			

Engine Failure	B1	May cause crash that ends in loss of life, environmental damage, and damage to the aircraft	Forced landing procedures, glide performance optimization, emergency landing sites mapping
Controlled Flight into Terrain	C1	May cause injury, loss of life, environmental damage, and damage to the aircraft	Terrain awareness systems, minimum altitude alerts, obstacle database, pilot training
Obstacle Collision	D1	Agricultural aircrafts prone to power line strikes and low altitude obstacles, can cause loss of control, aircraft damage, environmental damage, ground equipment damage, and injury/loss of life	Coordinated flight paths, enhanced pilot visibility training
Mid-Air Collision	E1	May cause crash that ends in loss of life, environmental damage, and damage to the aircraft	Air traffic monitoring system
High-G's	C2	Wing spar failures, fuselage structural, and structural failure at points of high stress	G-load monitoring, structural limits enforcement, pilots aware of load factors
Flight Control System Failure	B1	Can include elevator, aileron, or rudder system failures causing pitch, roll, or directional control loss	Redundant flight control systems, manual reversion capability, control authority limits.
Approach/Landing			
Hard Landing	B2	Can cause damage to fuselage, wings, and landing gear	Robust landing gear design, soft field landing procedures, pilot training
Uncontrolled Landing	B1	Pilot fatigue, speed errors, and wind conditions, can cause hard landing, damage to the aircraft, and it is landing environment	Enhanced approach speed procedures, go-around training, pilot cognition assessments
Vibration Wear	B3	Structural damage, control system wear leading to reduced control precision, and an offset engine alignment	Enhanced structural design with fatigue resistance materials, vibration dampening engine/avionics mounts, as well as schedule inspection intervals for those critical wear points

9. Structural Analysis

The complete CAD model was developed in Autodesk Fusion 360 based on the AT 802A and our design specifications. The model captures the overall geometry of the fuselage, wing planform, tail surfaces, and major aerodynamic features. The purpose of this model is to provide a reference for structural analysis and ensure the wing section analyzed in SolidWorks corresponds to the correct airfoil and span geometry.

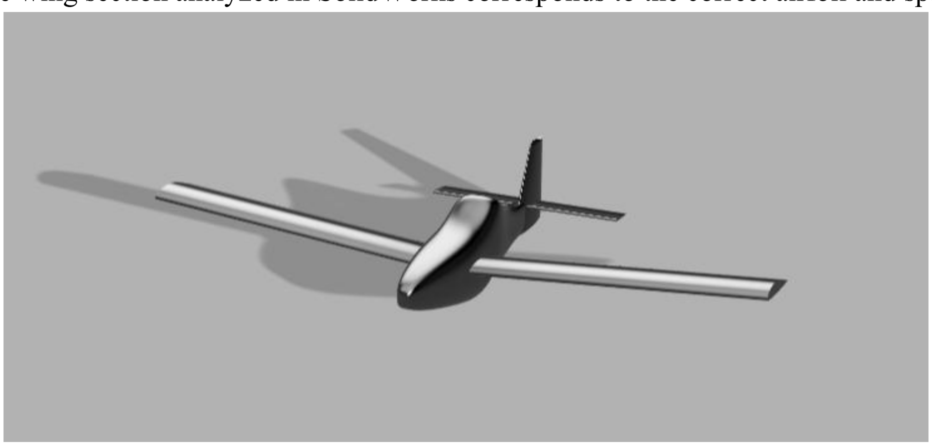


Figure 13: Sky Combine Model

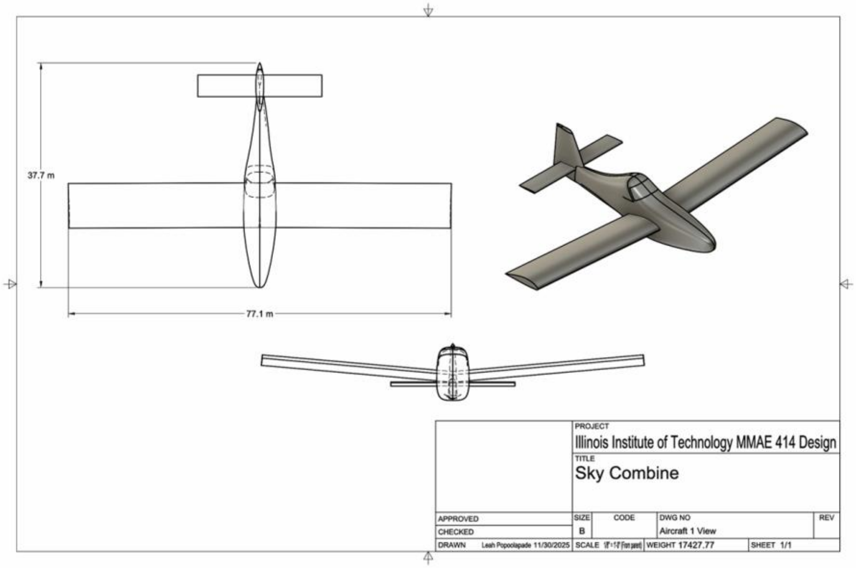


Figure 14: Sky Combine Drawing

The wing was modeled in SolidWorks using NACA 4415 airfoil cross sections, which provided moderate camber and thickness suitable for the agricultural aircraft mission profile of our design. The airfoil coordinates were imported into SolidWorks and extruded to represent the half-span of the wing, 11.75 meters. This geometry forms the structural simulation that allows for the evaluation of stress distribution, bending behavior and deflection under aerodynamic loading. For the structural evaluation, the wing and internal structure were assumed to be constructed from 2024-T3 aluminum alloy, which is commonly used due to its high strength-to-weight ratio.

To set up the simulation, a distributed force was applied upward on the bottom surface of the wing to represent the operating flight conditions. A fixed boundary condition was applied to the wing root to represent the attachment to the fuselage. This setup allows the simulation to capture bending stresses and overall deflection under realistic operating loads. The mesh was generated using tetrahedral elements with local refinement near the wing root where the highest stresses occur.

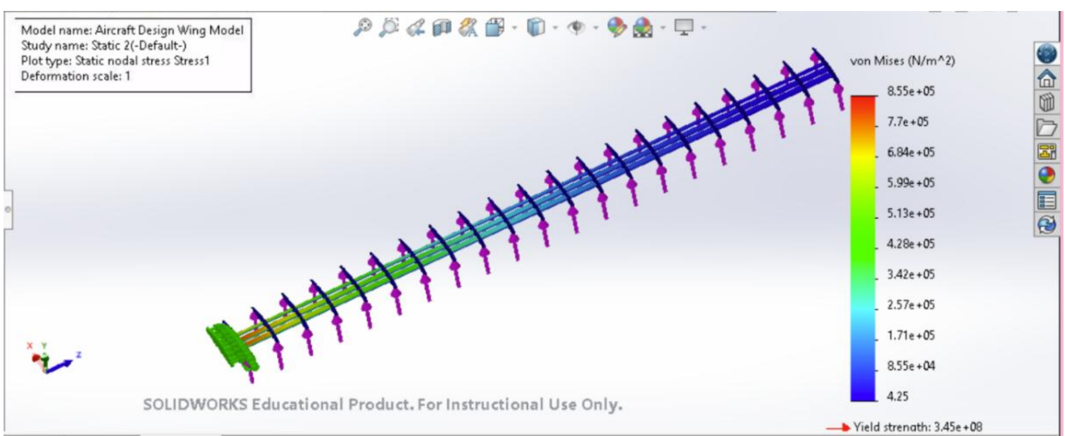


Figure 15: Wing Shear Stress

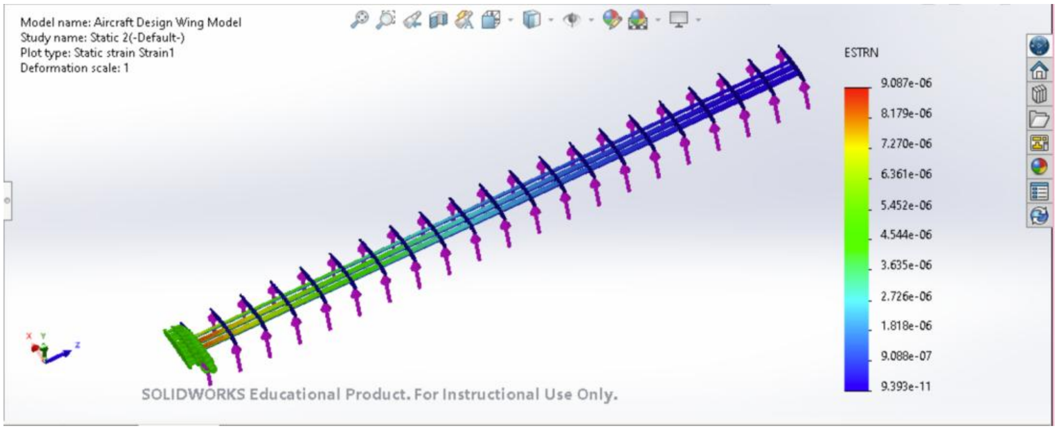


Figure 16: Wing Strain

The results indicate that the wing structure can withstand the aerodynamic loads without exceeding allowable stress limits. The stress concentrations observed near the wing root highlight the importance of reinforcement in this region. The deflection results are within an acceptable range for general aviation aircraft of this size. The low stress value comes from the simplified structural model, and the analysis represents only 1-g aerodynamic load. A real wing would experience higher stresses under maneuvering, but for our baseline load case the simplified model shows a large margin to yield, which confirms it is structurally safe.

The CAD model and structural simulation provide an understanding of the wing's performance under load. The results verify that the wing is structurally safe with large margin to yield. The use of Fusion 360 modeling and SolidWorks analysis ensures accurate representation of both geometry and structural behavior.

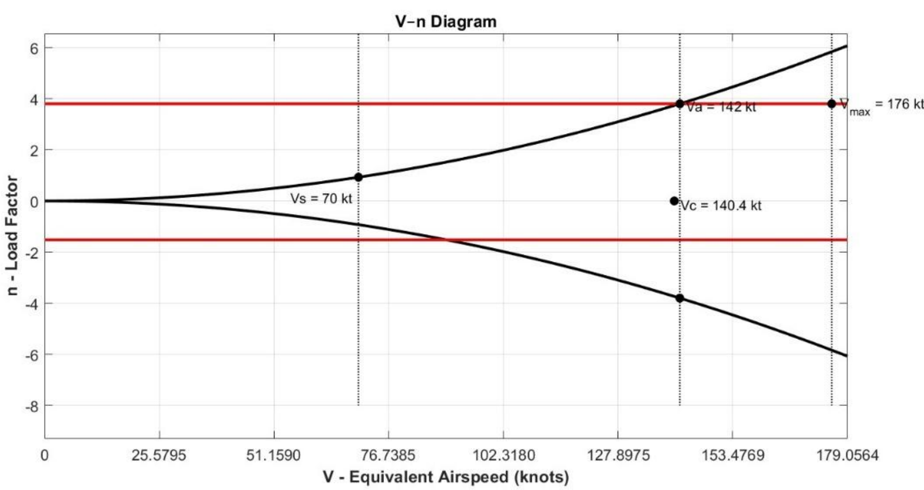


Figure 17: V-n Diagram

The V-n diagram defines the safe operating envelope of the aircraft by combining aerodynamic and structural limitations based on the NACA 4415 airfoil and the aircraft's weight. For this design, the stall speed is about 70 knots, and the stall-limited load factor increases with airspeed until it intersects the structural limit. The horizontal red lines represent the structural positive and negative load-factor limits of 3.8 and -2.2, respectively. The maneuvering speed Va = 142 knots. The cruise speed, Vc = 140.4 knots, lies below the maneuvering limit. Vmax represents the highest operating speed for which structural integrity is maintained. Beyond Vmax, small control deflections can produce loads that exceed the structural limits due to increased dynamic pressure. Overall, the V-n diagram shows that the aircraft has safe flight regions bound by stall at low speeds and structural limits at higher speeds.

Characteristic	Value
Maximum Takeoff Weight (lbf)	19000.0 (17427.77 gross weight)
Length (ft)	37.71
Wingspan (ft)	77.1
Wing Area (ft^2)	758.93
Wing Loading at Takeoff ($\frac{lbf^2}{ft}$)	23.97
Wing Loading at Cruise ($\frac{lbf^2}{ft}$)	22.96
Thrust-to-Weight Ratio at Takeoff	0.32
Thrust-to-Weight Ratio at Cruise	0.19
Wing Airfoil	NACA 4415
Aspect Ratio	7.83
Maximum Thickness (% chord)	15
Power-to-Weight Ratio (hp/lbf)	0.11
Propeller	5 blade/ 134” diameter
Engine	PT6A-68B
Range with Loiter (nmi)	600

Table 2: Optimized Design Characteristics

10. Conclusion

The Sky Combine has demonstrated that it can complete its intended mission profile. It can maintain a steady glide under trimmed conditions, closely matching the lift over drag ratio of more theoretical results. It can perform level flight, transition to stable climbing flight, and then return to level flight, all while staying under the maximum available thrust for the PT6A-68B. There were small instances of phugoid oscillation during both the initial climb and the transition to level flight, but these were quickly damped. It can hold a pattern for loitering, with small variations in its height across the entire flight profile. It can perform a series of bi-directional turns leading into glided landing; however, its final approach velocity was higher than desired. For all the simulations ran, the Sky Combine closely followed its commanded flight paths, a testament to its design. This was all possible because of the data gathered in XFLR5, as well as design decisions made regarding wing placement and sizing. The use of the 6-DOF Simulink model, along with the XFLR5 data, verified both longitudinal and lateral stability, while the structural analysis showed that the wing and airframe operate below the material limits and have large yield margins.

While the Sky Combine can complete its intended mission in its current form, there are areas that can be improved. The landing velocity is one thing that needs to be addressed not just for the safety of the aircraft, but for the safety of the pilot. Coming into land at 80 meters per second is dangerous and could easily result in the loss of control of the aircraft. More advanced thrust and elevator control in the Simulink model might be able to address this. The risk analysis also shows that several hazards tied to low-altitude operations and system failures still need attention, and while mitigation steps were identified, some of these risks would benefit from further refinement as the design develops. Even though the aircraft can complete its mission, the risk matrix makes it clear that improving reliability and reducing exposure to high-severity events should be part of the next stage of work.

11. References

[1] U.S. Department of Defense. MIL-STD-882E Department of Defense Standard Practice System Safety. Department of Defense, 27 Sept. 2023, https://www.ndeed.org/NDEEngineering/SafeDesign/MIL-STD-882E.pdf.

[2] U.S. Federal Aviation Administration. "14 CFR Part 137 - Agricultural Aircraft Operations." Electronic Code of Federal Regulations, Office of the Federal Register, 2024, www.ecfr.gov/current/title-14/chapter-I/subchapter-G/part-137.

[3] National Aeronautics and Space Administration. NASA System Safety Handbook, Volume 1: System Safety Framework and Concepts for Implementation. NASA, Nov. 2011, https://ntrs.nasa.gov/api/citations/20120003291/downloads/20120003291.pdf.

[4] "Aircraft Stability and Control Lecture 1." *MIT OpenCourseWare*, MIT, 2004, ocw.mit.edu/courses/16-333-aircraft-stability-and-control-fall-2004/9befd7ea465e31f9ff763c5587815264_lecture_1.pdf. Accessed 30 Nov. 2025.

[5] NACA 4415 (NACA4415-IL), airfoiltools.com/airfoil/details?airfoil=naca4415-il. Accessed 30 Nov. 2025.

[6] "UAV NACA4415 Wing Structural Performance Analysis Subjected to External Aerodynamic Load Using Schrenk's Approximation." Journal of Advanced Research in Fluid Mechanics and Thermal Sciences, Ernnie Illyani Basri, Adi Azriff Basri, Mohd. Firdaus Abas, Faizal Mustapha, Mohamed Thariq Hameed Sultan, Kamarul Arifin Ahmad, www.akademiabaru.com/doc/ARMSV50_N1_P19_25.pdf. Accessed 1 Dec. 2025.

[7] Raymer, Daniel P. *Aircraft Design : A Conceptual Approach*. 6th ed., Reston, Va, American Institute Of Aeronautics And Astronautics, Inc, 2018

12. Appendix A